
ECONOMIC AND FIELD GEOLOGY

Course Code : 332303

Programme Name/s : Mining & Mine Surveying/ Mining Engineering**Programme Code : MS/ MZ****Year : Second****Course Title : ECONOMIC AND FIELD GEOLOGY****Course Code : 332303****I. RATIONALE**

This course will help the students to use knowledge of geology, stratigraphy and remote sensing in mining engineering works, geological mapping, in search of ground water and nature of strata. After acquiring knowledge and related skill student will choose site according to its use for mining and civil engineering projects. This course is also useful for the ground control underground mines and slope stability in opencast mining

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help students to achieve following industry identified outcomes through various learning experiences in a) Use geological knowledge and skills to identify Indian geological formation b) Execute knowledge of an exploration and economic mineral, mining work by using conventional and remote sensing method.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Identify Ore minerals, Industrial minerals and Economical minerals.
- CO2 - Use knowledge of Geological Formation of India in given situation.
- CO3 - Apply prospecting and subsurface exploration method in given situation.
- CO4 - Illustrate Hydrological characteristics and interpretation of water table contour map.
- CO5 - Analyse Engineering Geological problems and study of remote sensing.

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Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
332303	ECONOMIC AND FIELD GEOLOGY	EFG	DSC	2	-	2	1	5	5	3	30	70	100	40	25	10	25#	10	50	20	200

Total IKS Hrs for Sem. : 1 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe given processes of Economic Minerals formation. TLO 1.2 Explain given properties of specified Economic Minerals. TLO 1.3 Describe given physical ,chemical properties and origin of Petroleum. TLO 1.4 Introduction to Minerals and their Exploitation in Ancient India.	Unit - I Economic Geology and Petroleum 1.1 Elements of Economic geology, Process of ore formation of economic Minerals deposits with examples. 1.2 Metaliferrous deposits of India – Iron,Copper,Mangnese,Aluminium,Gold,Lead and Zinc (Fe, Cu, Mg, Al, Au, Pb& Zn)Mineralogy, Origin, Distribution, uses, grades and reserves. 1.3 Petroleum-Physical and Chemical properties,Uses Origin and Occurrence of Petroleum in India. 1.4 Iron age in India.(1200BC- 600 AD) (IKS)	Video Demonstrations Presentations Lecture Using Chalk-Board

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Interpret principles of Stratigraphy and Physiographic Divisions of India. TLO 2.2 Describe given Geological Formations of India. TLO 2.3 Introduction to Ancient Geology.	Unit - II Stratigraphy 2.1 Principles of Stratigraphy, Geological Time-scale, Physiographic Divisions of India. 2.2 Major geological formations of India. Classification, Distribution, Lithology, Structures and Economic importance of following systems. 1) Archean & Dharwar system 2) Cuddapah system 3) Vindhyan System 4) Gondwana super group 5) Deccan trap 2.3 Minerals and Metals in Harappan era. (3200-1800BC) (IKS)	Demonstration Presentations Case Study Lecture Using Chalk-Board
3	TLO 3.1 Select relevant field equipments for Geological mapping. TLO 3.2 Describe various Geological Prospecting method. TLO 3.3 Use suitable Geological Prospecting method.	Unit - III Geological Field Work 3.1 Field equipment's required for geological Mapping, Method of field work, and collection of samples. 3.2 Geological prospecting-Airborne prospecting Method, aerial photographs, Ground prospecting Method, Surface prospecting Method, Mapping, Test pitting, trenching, editing, drilling. 3.3 Geophysical Methods of Prospecting -Magnetic, gravity, electrical, resistivity. 3.4 Geochemical Methods of prospecting-Soil, stream, water, rock sampling methods.	Case Study Demonstration Presentations Lecture Using Chalk-Board

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Describe Ground Water Hydrology. TLO 4.2 Explain various water bearing properties of rocks. TLO 4.3 Interpret water table Contour Map. TLO 4.4 Explain Groundwater Movement. TLO 4.5 Interpret Ground Water Provinces of India.	Unit - IV Ground Water Hydrology 4.1 Hydrological cycle, Zones of Ground water, Occurrence of ground water, Geological work done by groundwater. 4.2 Water bearing properties of rocks, Porosity, permeability. 4.3 Water table contour map. 4.4 Ground water monitoring system, Darcy's law of ground water movement, specific yield and specific retention, pumping of well. 4.5 Ground water provinces of India, Interpretation of map, location and types of rocks found in various provinces.	Video Demonstrations Presentations Lecture Using Chalk-Board

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Explain relevant engineering properties of rocks TLO 5.2 Investigate geological problems related to Dams and Tunnels in given situation. TLO 5.3 Describe Remote Sensing and GIS	Unit - V Engineering Geology and Remote sensing 5.1 Engineering properties of rocks ,Criteria for selection of building stones- availability, workability, strength, durability, porosity, permeability, texture, resistance the, colour and cost. 5.2 Geological problems associated with dam and tunnel side. 5.3 Introduction to remote sensing techniques ,analog and digital data product. 5.4 Remote sensing satellite and application of remote sensing in mining operation. 5.5 Introduction to GIS and its application.	Video Demonstrations Presentations Case Study Lecture Using Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify any two Iron ore minerals based on given physical properties.	1	*Identification of iron ore minerals (Hematite / Limonite / Magnetite/ Siderite/ Pyrite/ Pyrrhotite/ Goethite/ Siderite/ Marcasite.)	2	CO1
LLO 2.1 Identify any two Manganese ore minerals based on given physical properties.	2	*Identification Manganese ore minerals (Braunite / Pyrolusite/ Psilomelane/ Manganite.)	2	CO1

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Determine any two Iron ore minerals based on given physical properties.	3	*Identification of Copper ore minerals (Azurite/ Malachite/ Cuprite/ chalcopyrite/ covellite/ Bornite/ Chalcocite.)	2	CO1
LLO 4.1 identify of anyone mineral based on given physical properties.	4	*Identification Lead and Zinc ore minerals – (Galena, sphalerite)	2	CO1
LLO 5.1 identify any two Aluminium ore minerals based on given physical properties.	5	*Identification of Aluminium ore minerals (Bauxite/corundum/ gibbsite/Correndum/Diosper)	2	CO1
LLO 6.1 Interpret physical properties of any two Miscellaneous ore minerals.	6	*Identification of Miscellaneous ore minerals.(Chromite/ Ilmenite/ Baryte/ Magnesite/ Arsenopyrite/ Realgar/ Monazite/ Molybdenite/ Cinnabar/ Orpiment/ Wolframite.)	2	CO1
LLO 7.1 Select three suitable Industrial minerals based on given physical properties.	7	*Identification of Various types Industrial minerals (Talc/Kyanite, Graphite/Asbestos/Gypsum/Graphite/Coal/ Limestone/Calcite/Mica/Silliminite/Garnet/Asbestos.)	2	CO1
LLO 8.1 Draw Geological time scale with stratigraphical units.	8	*Indian Stratigraphy	2	CO2
LLO 9.1 Draw Geological succession of Archean and Gondwana rocks.	9	*Geological Formation of India.	2	CO2
LLO 10.1 Interpretation of topography from given topographical maps.	10	*Study of Toposheet (Any 5)	2	CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 11.1 Interpretation of Geology from given Satellite Imagery.	11	*Study of Satellite photographs. (Any three)	2	CO3
LLO 12.1 Draw outcrop map by three point problem method.	12	*Outcrop Map (Three point problem method)	2	CO3
LLO 13.1 Draw outcrop map when true dip is given on the map.	13	*Outcrop Map (True dip is given)	2	CO3
LLO 14.1 Draw outcrop map when out crop exposures are given on the map.	14	*Outcrop Map (Outcrop exposures are given)	2	CO3
LLO 15.1 Draw outcrop map when out crop exposures are given only on one contour.	15	*Outcrop Map (One Counter given)	2	CO3
LLO 16.1 Draw a outcrop map When out crop exposures are given in a bore hole on either side of fault.	16	*Outcrop Map(When out crop exposures are given in a bore hole on either side of fault.)	2	CO3
LLO 17.1 Calculation of strike,dip and depth of given strata,When out crop exposures are given in a bore hole on either side of fault.	17	*Calculation of strike,dip and depth of given strata.	2	CO3
LLO 18.1 Calculate dip and strike of beds by actual measurement in the field by Compass.	18	*Aplication of Brunton/Clinometer Compass.	2	CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 19.1 Calculate apparent dip when true dip is given in amount by graphical method.	19	Calculation of apperent dip when true dip is given .	2	CO3
LLO 20.1 Calculate true dip when apparent dip is given in amount by graphical method.	20	Calculation of true dip when apparent dip is given.	2	CO3
LLO 21.1 Draw a water table map on the basis of given water table data of the area.	21	Water Table Map (By given water table data)	2	CO4
LLO 22.1 Interpretation of Remote Sensing Techniques in ground water survey.	22	Remote Sensing Techniques in ground water survey.	2	CO4
LLO 23.1 Identify various Engineering properties of rocks such as, Porosity, Permeability, Strength, Hardness.	23	Engineering Properties of Rocks .	2	CO4
LLO 24.1 Prepare a report on any one dam sites of Indian dam project.	24	Dam Sites Visit on Indian dam project.	2	CO5
LLO 25.1 Acquire knowledge about ground water monitoring.	25	Ground Water Monitoring	2	CO4

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- 1)Collect photographs and samples of ore minerals by geological field work and prepare report. 2)Interpret the geological succession of nearby area prepare a report. 3)Visit to nearby site of G.S.I. and submit report of prospecting exploration work of the site you visited. 4)Visit to remote sensing station. Collect satellite imaginaries of the earth. 5)Visit to hydrological survey of India or G.S.D.A. submit report of ground water exploration work,photographs. Prepare rain water harvesting model or “jalyuktshihar yojna”. 6)Visit to nearby dam or road, tunnel or bridge site, check how the stability of structure is checked, collect the samples, photographs and prepare report.

Assignment

- Prepare chart on Geological Time Scale and Geological succession. Indian mining industry in respect of any one metallic deposit. (Iron/Copper/Manganese/Cement/Coal) Prepare chart on Ground Water zone. Prepare report on Geological Field work by visiting near by Geological Camp. Prepare a assignment on Remote Sensing Satellite.

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- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Set of Iron ore minerals.(magnifying lenses, pen knife, streak plate)	1
2	Note-From Sr.No.1-7 must contain at least 2 or 3 sets of minerals,ores and each set contain at least 6 different samples.	1,2,3,4,5,6,7
3	Toposheet of survey of India 4x4 feet (5 Toposheet)	10
4	Colour Satellite images between 2 and 5 meter high resolution.(5 Images)	11,22
5	Set of outcrop maps with mentioned details.(set squares, roller scale, and protractor.)	12,13,14,15,16,17
6	Brunton compass and Clinometer compass (damping time of needle: <14 seconds dial scale : 1° size : 80 x 70 x 35 (mm) weight : 250 grams.), set squares, roller scale, and protractor.	18,19,20
7	Set of Manganese ore minerals.(magnifying lenses, pen knife, streak plate)	2

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Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
8	Bore hole data of water level.	21
9	UTM, Shear Stress, Compression test.	23
10	Set of Copper ore minerals. (magnifying lenses, pen knife, strike plate)	3
11	Set of Lead and Zinc ore minerals (magnifying lenses, pen knife, streak plate)	4
12	Set of Aluminium ore minerals. (magnifying lenses, pen knife, streak plate)	5
13	Set of Miscellaneous ore minerals. (magnifying lenses, pen knife and streak plates)	6
14	Set Industrial Minerals. (magnifying lenses, pen knife, streak plate)	7
15	Relevant chart and books.	8,9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Economic Geology and Petroleum	CO1	12	2	4	8	14
2	II	Stratigraphy	CO2	14	6	4	6	16
3	III	Geological Field Work	CO3	12	2	6	6	14
4	IV	Ground Water Hydrology	CO4	10	4	4	6	14
5	V	Engineering Geology and Remote sensing	CO5	12	2	4	6	12
Grand Total				60	16	22	32	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)****MSBTE Approval Dt. 01/10/2024****Semester - 2, K Scheme**

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- Mid term Test,Rubrics for CO's,Assignment,Self Learning and Term work,Seminar /presentation.

Summative Assessment (Assessment of Learning)

- Lab performance,End of term examination,viva voce
- Lab performance,End of term examination,viva voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3			2	2	1	3			
CO2	3			1	2	1	3			
CO3	3	2		2	3	3	3			
CO4	3	3		2	3	3	3			
CO5	3	3		2	3	3	3			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

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Sr.No	Author	Title	Publisher with ISBN Number
1	Parbeen Singh	Engineering and General Geology	S.K. Kataria and Sons, 2013, ISBN-13: 978-9350142677
2	Bangar K.M	Principles of Engineering Geology	Standard Publishers Distributors Delhi, 2013, ISBN-13: 978-8180141157
3	DR.S.KUMAR	BASICS OF REMOTE SENSING AND GIS	UNIVERSITY SCIENCE PRESS ISBN NO.978-81-318-0544-2
4	R.N.P.AROGYASWAMI	COURSES IN MINING GEOLOGY	CBS PUBLISHER ISBN 13-978-8120409378
5	Dr.R.K.Bopche Dr.D.K.Agarwal	General and Engineering Geology	Global Education Limited 2018 ISBN-978-81-935799-2-3
6	A.PARTHSARATHI,V. PANCHAPAKESAN,R. NAGARJUN	ENGINEERING GEOLOGY	WILEY V. B. PUBLICATION NEW DELHI 2003 ISBN 978-8126509461
7	G. B. Mahapatra	A Textbook of Physical Geology	CBS Publishers & Distributors, ISBN 10-8123901100

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.gsi.gov.in/webcenter/portal/OCBIS	It has reports, maps, manuals and Guidelines. Training programmes for learning
2	https://pubs.geoscienceworld.org/	a free, public map-based toolset that allows users to search for cross sections, charts, tables, fig

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Sr.No	Link / Portal	Description
3	www.geologyshop.com	printed products including maps, sheet explanations and regional guides.
4	www.geologyshop.com	printed products including maps, sheet explanations and regional guides.
5	www.geology.com	This web site has information and News about Geology and Earth Science
6	https://www.surveyofindia.gov.in/	It is a official website of Survey of India having information about maps, mines and geological info
7	https://www.bgs.ac.uk/discovering-geology/	This is good for enhancing knowledge by watching few relevant programmes on Mines and Mine Survey
8	https://youtube.com/@ExploringGeologybyGeolovers	This video is good on Exploring Geology in Hindi and information about Jobs in India.
9	https://www.mecl.co.in/Index.aspx	Site related to exploration related work.
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

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